

Question 1

Sample Java program to check whether the given number is prime palindrome or not

Step wise execution

1. The program starts execution.
2. The Scanner object is created for taking input.
3. The user enters the number 131.
4. The entered number is stored as the original number.
5. The program starts reversing the entered number.
6. The last digit is taken one by one and added to the reversed number.
7. After all the digits are processed, the reversed number obtained is 131.
8. The reversed number is displayed on the screen.
9. The original number 131 is compared with the reversed number 131.
10. Both numbers are equal, so the given number is a palindrome.
11. The program then checks whether the palindrome number is prime.
12. The number 131 is checked for divisibility by numbers starting from 2.
13. The number 131 is not exactly divisible by any number other than 1 and itself.
14. Therefore, 131 is a prime number.
15. Since 131 is both prime and palindrome, the message
 "The number is prime palindrome" is displayed.
16. The program ends.

SAMPLE OUTPUT:

enter n:131

Reversed number: 131

The number is prime palindrome

Algorithm

Step 1: START

Step 2: Read the integer n.

Step 3: Set result = 0.

Step 4: Set original = n.

Step 5: Repeat while n != 0:

$r = n \% 10$

$n = n / 10$

$result = result * 10 + r$

Step 6: Display the reversed number.

Step 7: Set prime = 0.

Step 8: If original == result, then:

 a. Repeat for i = 2 to result - 1:

 If result % i == 0, then:

 Display "The number is not prime".

 Set prime = 0.

 Break the loop.

 Otherwise:

 Set prime = 1.

 b. If prime == 1, then:

 Display "The number is prime palindrome".

c. Otherwise, if $\text{prime} == 0$ and $\text{original} == \text{result}$:

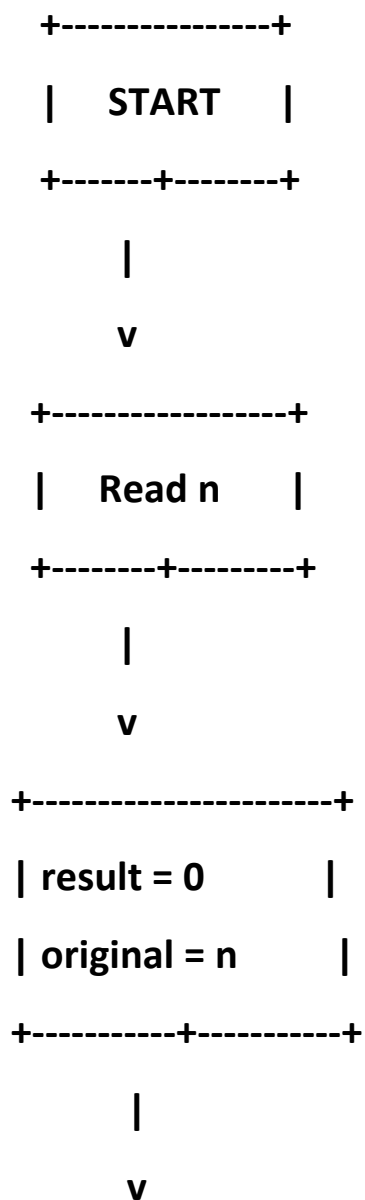
Display "The number is a palindrome".

Step 9: Otherwise:

Display "The number is not a palindrome".

Step 10: STOP

UML diagram



```

+-----+
| n != 0? |
+-----+
Yes| |No
  | |
  v |
+-----+
| r = n % 10 |
| n = n / 10 |
| result=result*10 |
|   + r   |
+-----+
|
|
+-----> n != 0?
      |
      v
+-----+
| Print Reversed |
| Number      |
+-----+
|

```

v

+-----+

| original == |

| result? |

+-----+-----+

Yes | | No

| |

| +-----+

|

|

v

v

+-----+ +-----+

| prime = 0 | | Print "Not a |

+-----+-----+ | palindrome" |

| +-----+-----+

v

|

+-----+ |

| i = 2 | |

+-----+-----+ |

|

|

v

|

+-----+ |

| i < result? | |

```

+-----+-----+      |
    Yes | | No      |
        | |          |
        v v          |

+-----+ +-----+ |
| result % i | | prime == 1? | |
| == 0?    | +-----+ |
+-----+ +-----+ |
    Yes | | No      |
        | |          |
        v v      v   |

+-----+ +-----+ |
| Not Prime | | Prime   | |
| prime = 0 | | Palindrome | |
| BREAK    | +-----+ |
+-----+ +-----+ |
        |          |
        +-----+ |
            |      |
            v      |

+-----+ +-----+ |
| Palindrome | |      |

```

```

| if not prime |   |
+-----+-----+   |
      |           |
      +-----+-----+
          |
          v
      +-----+
      |  STOP  |
      +-----+

```

Code:

```

package basics;
import java.util.Scanner;
public class Palindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("enter n:");
        int n = sc.nextInt();
        int result=0,original=n;
        while(n!=0){
            int r=n%10;
            n=n/10;
            result=result*10+r;
        }
        System.out.println("Reversed number: " + result);
    }
}

```

```
int prime=0;
if(original==result){
    for(int i=2;i<result;i++){
        if(result%i==0){
            System.out.println("The number is not
prime");
            prime=0;
            break;
        }prime=1;
    }
    if(prime==1){
        System.out.println("The number is prime
palindrome");
    }else if(prime==0 && original==result){
        System.out.println("The number is a palindrome");
    }else{
        System.out.println("The number is not a
palindrome");
    }
}
}
```

Output screen shot

```
File Edit Selection View Go Run Terminal Help
debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.Palindrome'
enter n:323
Reversed number: 323
The number is not prime
The number is a palindrome
PS C:\Users\Admin\OneDrive\Documents\basicjava> 121
PS C:\Users\Admin\OneDrive\Documents\basicjava> c;; cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.Palindrome'
enter n:121
Reversed number: 121
The number is not prime
The number is a palindrome
PS C:\Users\Admin\OneDrive\Documents\basicjava> c;; cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.Palindrome'
enter n:191
Reversed number: 191
The number is prime palindrome
PS C:\Users\Admin\OneDrive\Documents\basicjava> c;; cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.Palindrome'
enter n:181
Reversed number: 181
The number is prime palindrome
PS C:\Users\Admin\OneDrive\Documents\basicjava> c;; cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.Palindrome'
enter n:767
Reversed number: 767
The number is not prime
The number is a palindrome
PS C:\Users\Admin\OneDrive\Documents\basicjava> c;; cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.Palindrome'
enter n:3297923
Reversed number: 2297923
The number is not prime
The number is a palindrome
PS C:\Users\Admin\OneDrive\Documents\basicjava> c;; cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.Palindrome'
enter n:2332
Reversed number: 2332
The number is not prime
The number is a palindrome
```

Question 2

Write a java program which reads marks of five subjects through command line and print the total and average.

Step wise execution

Example Input:

Array size = 5

Array elements = 10 20 30 40 50

1. The program starts execution.
2. A Scanner object is created to take input from the user.
3. The user enters the size of the array as 5.
4. An integer array of size 5 is created.

5. The total is initialized to 0.
6. The program asks the user to enter 5 elements.
7. The elements are entered one by one and added to the total.
8. After entering all the elements, the array contains:
10 20 30 40 50
9. The total of all the elements is calculated as 150.
10. The average is calculated as $150 / 5 = 30$.
11. The program displays all the elements of the array.
12. The total value 150 is displayed.
13. The average value 30 is displayed.
14. The program ends.

SAMPLE OUTPUT:

enter array size:5

Enter 5 elements of the array:

10

20

30

40

50

10

20

30

40

50

Total: 150

Average: 30

Algorithm

Step 1: START

Step 2: Read the size of the array.

Step 3: Create an integer array with the given size.

Step 4: Set total = 0.

Step 5: Read the elements of the array one by one.

Step 6: Add each element to total.

Step 7: Calculate the average using:

$$\text{average} = \text{total} / \text{array size}$$

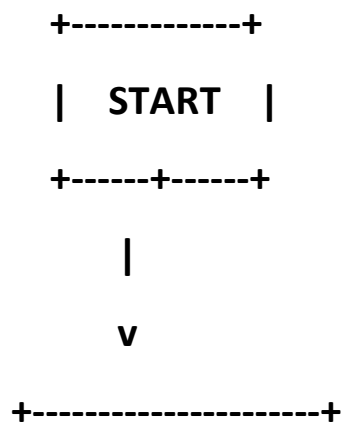
Step 8: Display all the elements of the array.

Step 9: Display the total.

Step 10: Display the average.

Step 11: STOP

UML diagram



| Read array size |

+-----+-----+

|

v

+-----+-----+

| Create integer array |

+-----+-----+

|

v

+-----+-----+

| Set total = 0 |

+-----+-----+

|

v

+-----+

| Read array |

| elements |

+-----+-----+

|

v

+-----+

| Add each |

| element to |

| total |

+-----+-----+

|

v

+-----+

| Calculate average |

| average = total / |

| array length |

+-----+-----+

|

v

+-----+

| Display array |

| elements |

+-----+-----+

|

v

+-----+

| Display total |

+-----+-----+

|

```

      v
+-----+
| Display average |
+-----+-----+
      |
      v
+-----+
|  STOP  |
+-----+

```

Code

```

package basics;
import java.util.Scanner;
public class ArrayAvg {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("enter array size:");
        int size=sc.nextInt();
        int[] arr=new int[size];
        int total=0;
        System.out.println("Enter " + size + " elements of the
array:");
        for(int i=0;i<arr.length;i++){
            arr[i]=sc.nextInt();
            total=total+arr[i];
        }int avg=total/arr.length;
        for(int x:arr){
            System.out.println(x);
        }
    }
}

```

```

        System.out.println("Total: " + total);
        System.out.println("Average: " + avg);
    }
}

```

Output screen shot

```

debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.ArrayAvg'
enter array size:5
Enter 5 elements of the array:
100
98
98
89
78
100
98
98
89
78
Total: 463
Average: 92
PS C:\Users\Admin\OneDrive\Documents\basicjava> c:: cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
debtailsInExceptionMessages' '-cp' 'C:\Users\Admin\OneDrive\Documents\basicjava\bin' 'basics.ArrayAvg'
enter array size:3
Enter 3 elements of the array:
40
45
50
40
45
50
Total: 135
Average: 45
PS C:\Users\Admin\OneDrive\Documents\basicjava> c:: cd 'C:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo

```